## Exercise path

Start with exercises 1 and 2. Then choose one of exercises 3 or 4 and do it without help. Exercises 5 to 7 are a reserve for transfer, investigation or programming: the teacher designates one. Exercise 8 is revisited after a delay.

### Exercise 8.1 · A distribution to build

A fair six-sided die, with faces numbered 1 to 6, is rolled. The random variable $X$ associates with the result the remainder of its Euclidean division by 3. For example, a result divisible by 3 gives $X = 0$.

1. For each of the six faces, state the value taken by $X$.
2. Present the probability distribution of $X$ in a table and check that the probabilities sum to 1.
3. Calculate $P(X < 2)$, stating which values of $X$ are retained.

### Exercise 8.2 · Two variance calculations

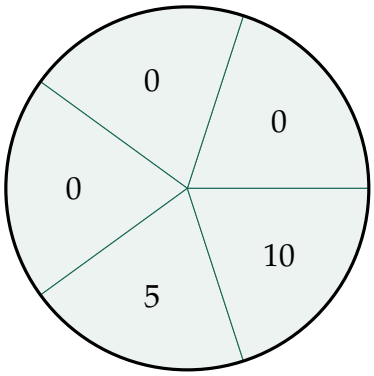
The random variable $X$ takes the values $-2$, 0 and 4 with respective probabilities $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{1}{4}$.

1. Calculate its expectation $\mu = E(X)$ and the second moment $E(X^2)$.
2. Calculate $V(X)$ using the König-Huygens formula, then deduce the exact standard deviation.
3. Recover the variance from the definition $V(X) = E((X - \mu)^2)$. Present the three contributions to the calculation.

### Exercise 8.3 · A fair game

A wheel has five sectors of equal angle, bearing the numbers 0, 0, 0, 5 and 10. The five sectors are assumed equally likely. The number obtained represents a gross gain $G$, in monetary units. To play, one pays a stake $m$; the net gain is therefore $X = G - m$.

1. Establish the distribution of $G$, then determine the stake $m$ that makes the game fair, that is, such that $E(X) = 0$.
2. After studying variance, calculate the standard deviation of the gross gain. Deduce that of the net gain when the stake is fair.
3. Explain why changing the stake changes the expectation of the net gain but not its standard deviation.



The five sectors have the same angle. The three sectors marked 0 are distinct and must all be counted.

### Exercise 8.4 · Same mean, different risk

Two fictitious investments are offered. The first yields a certain gain $X = 1$. The second yields a gain $Y = -2$ with probability $\frac{1}{2}$ and a gain $Y = 4$ with probability $\frac{1}{2}$. The gains are expressed in the same monetary unit.

1. Compare the expectations of the two gains, then their variances.
2. Someone states: "These investments are equivalent since they have the same mean." Explain what this sentence omits and what the variance makes it possible to distinguish here.

### Exercise 8.5 · Linearity without independence

A fair die numbered 1 to 6 is rolled. Let $X$ be the result of the roll and define $Y = 7 - X$. The two variables are therefore obtained from the same roll.

1. Calculate $E(X)$ and $E(Y)$, then $E(X + Y)$. Check the linearity of expectation on this example.
2. Are the variables $X$ and $Y$ independent? Justify using, for example, the events $\{X = 1\}$ and $\{Y = 6\}$.
3. Does the linearity of expectation require independence? Conclude using your results.

### Exercise 8.6 · A minimal loss

A random variable $X$ takes finitely many values and has an expectation $\mu$. For a real number $a$, the mean squared error associated with the constant forecast $a$ is measured by $E((X - a)^2)$.

1. Prove that $E((X - a)^2) = V(X) + (a - \mu)^2$. You may write $X - a = (X - \mu) + (\mu - a)$.
2. Deduce the value of $a$ that minimises this error, then the minimum value. Justify the uniqueness of the optimal choice.

### Exercise 8.7 · Letter frequencies

We wish to measure the frequency of the letter "a" in a text. The denominator is the number of alphabetic characters: spaces, digits and punctuation marks are excluded. Capital letters are treated as lower case; accented letters remain distinct from "a".

1. Write a Python function that returns this frequency. State the result chosen when the text contains no letters.
2. Test the function with `Aba !`, then with `123 !` and with the empty string. Explain the expected results before running the program.

### Exercise 8.8 · Delayed review

An experiment whose probability of success is $\frac{1}{2}$ is repeated twice, independently. The random variable $X$ counts the number of successes obtained.

1. Recover the distribution of $X$ using the four possible sequences of results.
2. Calculate $E(X)$ and $V(X)$ starting from this distribution.

**Homework.** Do only the task designated at the end of the session, in the dated progression: 10–20 min at most. The exercises depend on the topics already studied, so some questions are deferred. Revisit exercise 8 one week after the chapter (10 min). If the allotted time is exceeded, note the difficulty and bring your attempts; do not pile up the whole reserve.

## To do in the CQFD textbook

These exercises are done in your textbook. The progression indicates the session; they replace the homework planned for that day (10 to 20 minutes in total).

**CQFD, exercise 28 p. 306, all questions.** Calculate the expectation from the distribution table; keep the probabilities associated with each value.

**CQFD, exercise 31 p. 306, all questions; use the distribution of exercise 28.** Calculate the variance and then the standard deviation, rounding to one decimal place as requested in the textbook.